DAY 3 · FINANCE

Are we bleeding cash on interstate freight?

> *"GMV is growing, but freight spend is running at an alarming R$ 466K/quarter. We have complaints about delivery fees exceeding product prices. Are we bleeding cash on interstate logistics? Audit our corridor economics before the board meeting."*
>
> **Henrique Silva**, Chief Financial Officer

46,219
order items analysed

R$ 927K
freight charged in six months

27
customer states compared

Prepared by Shatayu Roy · MySQL analysis of the public Olist dataset, January to June 2018 · Simulated case

THE BRIEF

What needed to be checked

Henrique’s questions

- Is freight growing faster than sales?
- How often does delivery cost more than the item?
- Are we losing cash on deliveries between states?
- Which routes drive the cost?

My own checks

- Who actually pays the freight?
- Is the interstate gap real, or just pricier items?
- Why do some regions pay so much more?
- Which items had no freight at all?

THE ANSWER

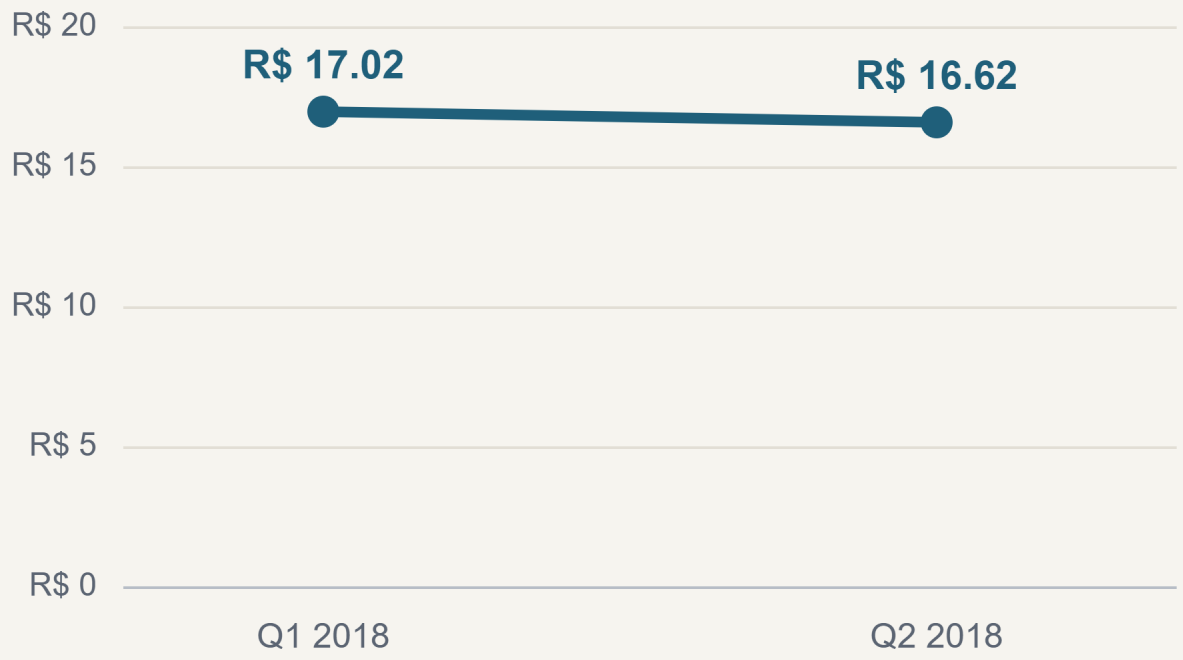
Customers pay the freight in full; the open problem is the Northeast burden

1 · GROWTH	Freight per R$ 100 of goods fell from R$ 17.02 to R$ 16.62.
2 · WHO PAYS	In 99.76% of orders the customer paid price plus freight.
3 · INTERSTATE	Interstate items pay R$ 18.92 per R$ 100; local items R$ 12.28.
4 · ROUTES	Ten routes carry 52% of interstate freight; SP to RJ and MG alone 22%.
5 · LOCAL SUPPLY	Northeast buyers have almost no local sellers and pay R$ 21 to R$ 28.

1 · GROWTH

Freight is not outpacing sales: it fell from R$ 17.02 to R$ 16.62 per R$ 100

Freight charged for every R$ 100 of goods sold



+3.8% vs +1.4%

GMV growth vs freight growth, Q1 to Q2 2018. Sales grew almost three times faster.

R$ 466,475

Q2 freight. Henrique's figure is right, but it is stable (Q1: R$ 460,216).

Delivered items grouped by purchase date · Step 1 · Olist delivery audit · Day 3

2 · WHO PAYS

Customers pay the freight in 99.76% of orders

ITEM PRICE
R$ 5.51M

+

FREIGHT
R$ 0.93M

=

THE BILL
R$ 6.44M

≈

CUSTOMERS PAID
R$ 6.44M

40,177 of 40,273

orders matched within R$ 0.05. The other 96 paid more, never less, probably instalment interest.

Not in the data

What Olist pays the carriers. Whether Olist loses money on the carrier side stays open.

3 · INTERSTATE

Interstate items carry 75% of freight and most fee complaints

Q2 2018, delivered items

MEASURE	LOCAL	INTERSTATE
Share of items	39.7%	60.3%
Average freight per item	R$ 13.28	R$ 25.41
Freight per R$ 100 of goods	R$ 12.28	R$ 18.92
Freight equal to or above price	1.13%	6.03%

90%

of items with freight equal to or above price were interstate (1,593 of 1,769 in H1). Yet only about 6 in 10 items are interstate.

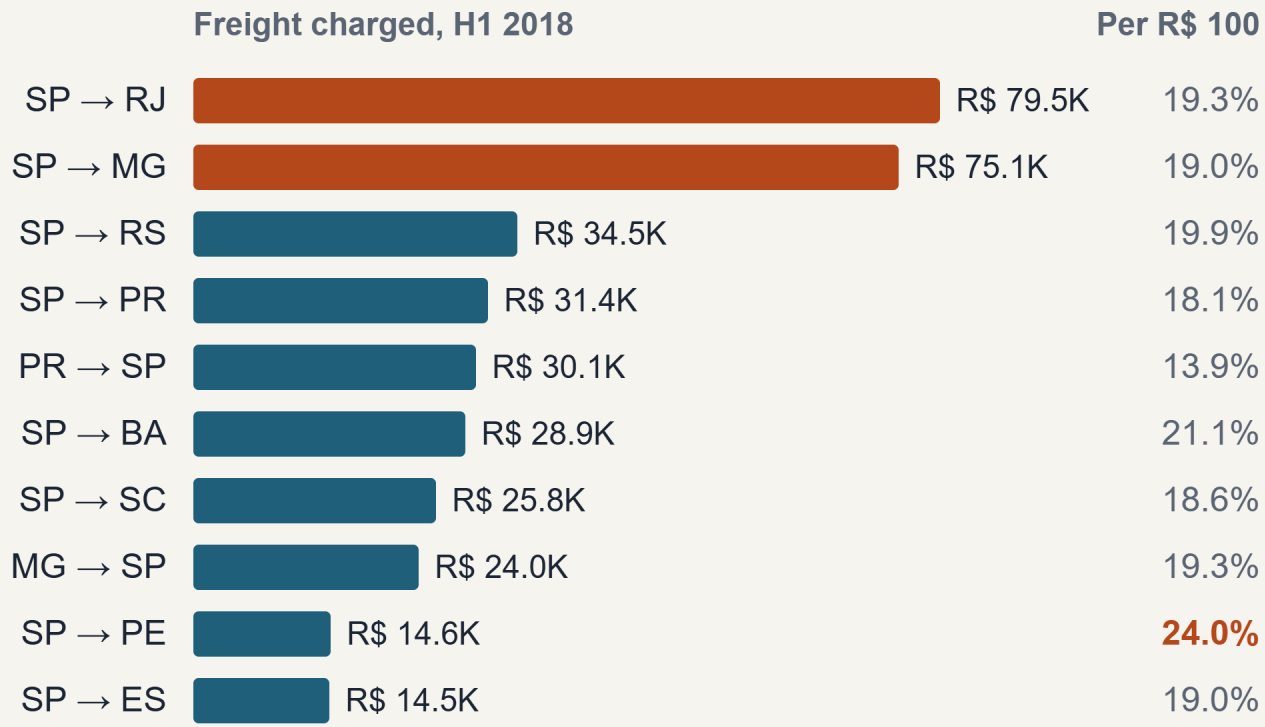
R$ 43K

freight on those items, 4.7% of H1 freight: a complaint problem, not a cash problem.

Interstate = seller and customer in different states · Steps 3, 4 and 5 · Olist delivery audit · Day 3

4 · ROUTES

Ten routes carry 52% of interstate freight; eight start in São Paulo



22%

of interstate freight on SP to RJ and SP to MG alone, at a normal R$ 19 per R$ 100. Big volume, normal burden.

24%

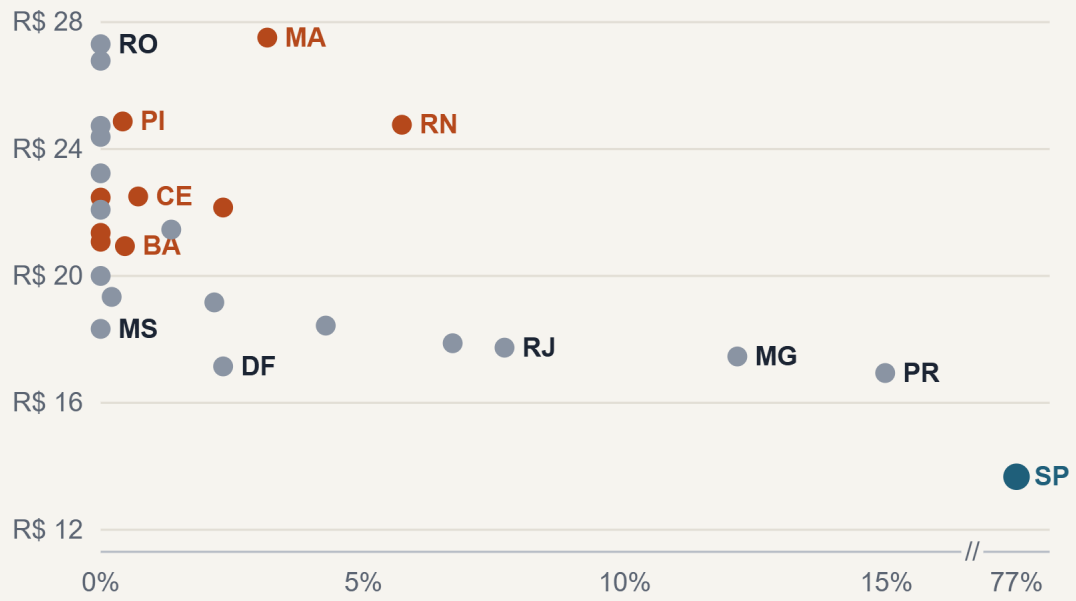
on SP to PE, the heaviest in the top 10. Far routes have small volume but a heavy burden.

Routes with at least 20 items · Step 6 · Olist delivery audit · Day 3

5 · LOCAL SUPPLY

Where buyers have no local sellers, delivery costs the most

Each dot is a customer state: freight per R$ 100 (up) vs share bought locally (right)



● Northeast states · ● São Paulo · ● other states

H1 2018 delivered items, by customer state · Step 7 · Olist delivery audit · Day 3

7 of 1,510

items Bahia's customers bought came from Bahia sellers. The rest travelled from other states.

Two causes, mixed together

The Northeast is also far from São Paulo. Distance was not measured here, so the two effects are not split.

WHAT TO DO

Recommendations

0

Get carrier invoices first

The dataset has no carrier-cost column. Invoices answer the margin question and show where rates are too high.

From finding 2

0.46%

Recruit Northeast sellers

Start with Bahia. Local items pay R$ 12.28 per R$ 100, interstate R$ 18.92. A warehouse only once volume grows.

From findings 3 and 5

22%

Then negotiate busy routes

If invoices show room, start with SP to RJ and SP to MG: 22% of interstate freight. Finance decides who keeps the saving.

From finding 4

Success measure: in Bahia next half-year, local purchase share and freight per R$ 100, against 0.46% and R$ 20.95 in H1 2018.

WHY IT MATTERS

What the company gains from acting on freight

R$ 7.4K

Per R$ 1 cut, per half-year

SP to RJ and SP to MG moved 7,415 items in H1. Every R$ 1 cut per item there is worth about R$ 7.4K, to customers or to Olist. A scenario to size the talks.

4,013

Northeast items to bring closer

Bought by Northeast customers in H1, almost all shipped from other states at R$ 21 to R$ 28 per R$ 100. The biggest gap local sellers can close.

4.7%

No false alarm at the board

Items with fees above price hold only R$ 43K of freight. The board can treat them as a customer issue, not a cash emergency.

Not measured here: lost sales from high fees, and what Olist pays carriers. Carrier invoices would put a margin figure on this.

NOTES

Data notes and limitations

+11%

rise in average interstate freight, but per R$ 100 it barely moved (18.69 to 18.92): pricier items, not higher rates.

368

items (0.8%) had zero freight, likely free-shipping offers. Who paid the carrier is not in the data.

Items

counted, not orders, because each item has its own seller state. Orders hold 1.14 items on average.

What could change the picture

- Carrier cost and lost sales are not in the data.
- Two quarters only: a direction, not a long trend.
- Routes are state to state, not distance; neighbouring states count as interstate.
- Fees above price are counted per item; per order the share would be slightly lower.

NEXT · DAY 4 · GROWTH & FINTECH

Are we bleeding acquisition money into a leaky bucket?

Day 3 found that customers pay the freight, and the heaviest burden falls on Northeast buyers with no local sellers. The next morning, Growth asks a different question: do the customers we win ever come back?

> *"Marketing is demanding another R$ 2M acquisition budget. But Boleto orders take over a day just to approve, and rumor has it almost nobody ever buys from us a second time. Are we bleeding CAC into a leaky bucket?"*
>
> **Lucas Rocha**, Head of Growth & FinTech

SQL queries, results and this deck: **github.com/shatayuroy77-netizen/olist-logistics-margin-audit**